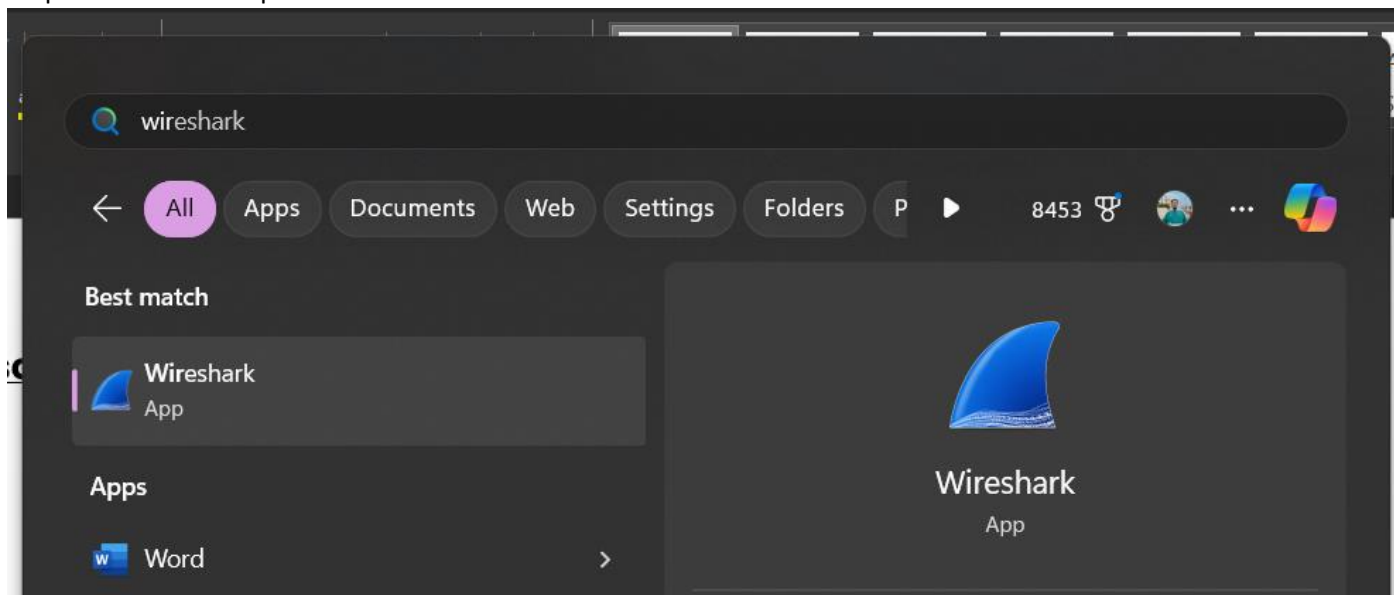
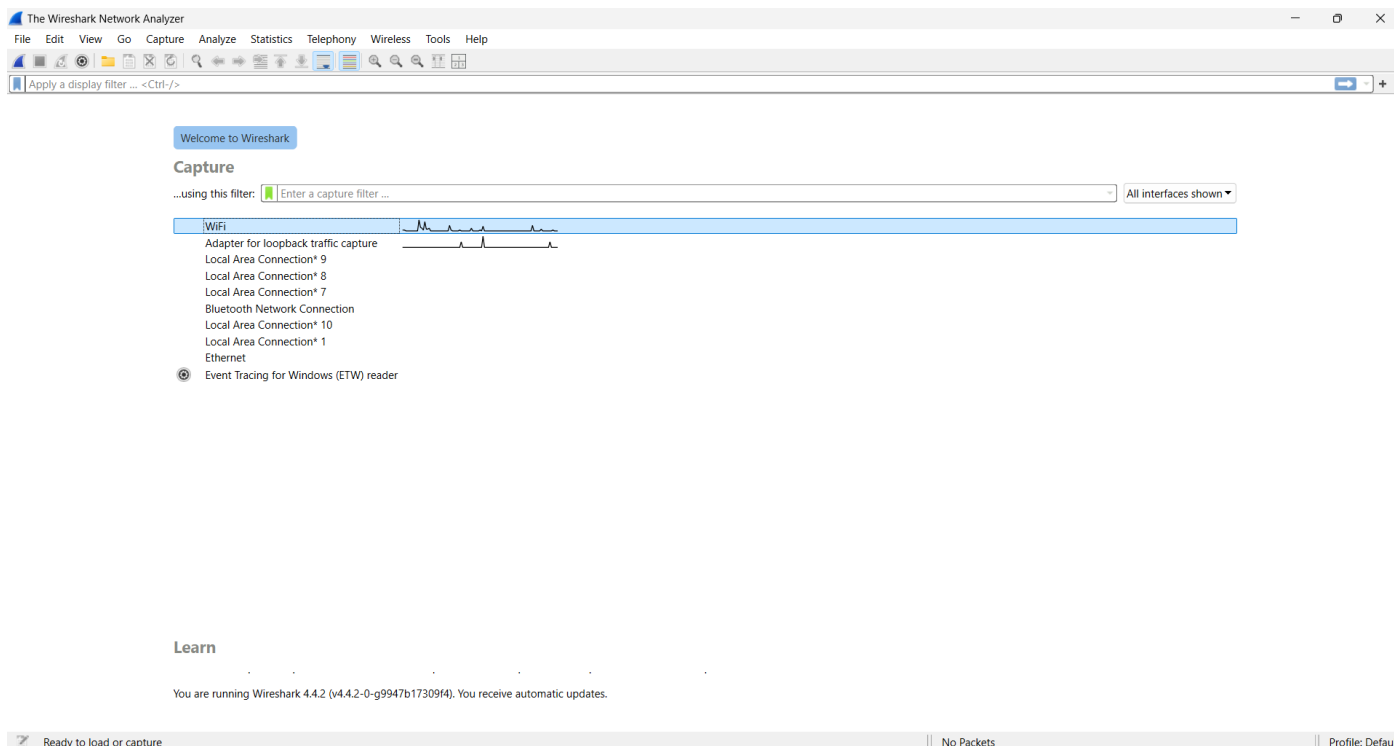


Capture some packets and analyze the header using Wireshark

Step 1 : Search and Open Wireshark



Step 2 : Select one of the interface,



Step 3 : Monitor and Start capturing the packets,

The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The main display area is divided into three panes: the top pane shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info; the middle pane shows the packet details for the selected packet (No. 1395, Time 143.266349); and the bottom pane shows the raw packet data in hexadecimal and ASCII. The selected packet is an ICMPv6 Neighbor Solicitation from 2400:1a00:b012:9774:946e:175b:dd1d:393d to 2400:1a00:b012:9774:1c2b:5024:f805:e99c. The packet details pane shows the ICMPv6 header and the Neighbor Solicitation message. The raw packet data pane shows the hexadecimal and ASCII representation of the packet data.

Step 4 : Checking IP address of website,

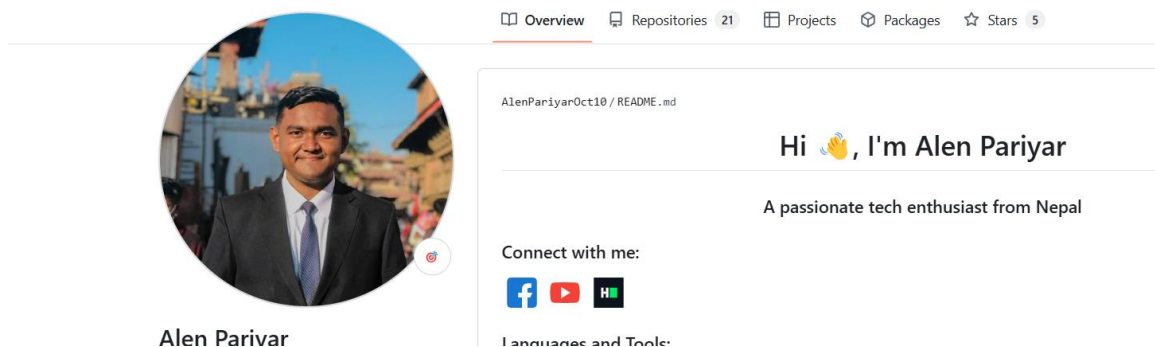
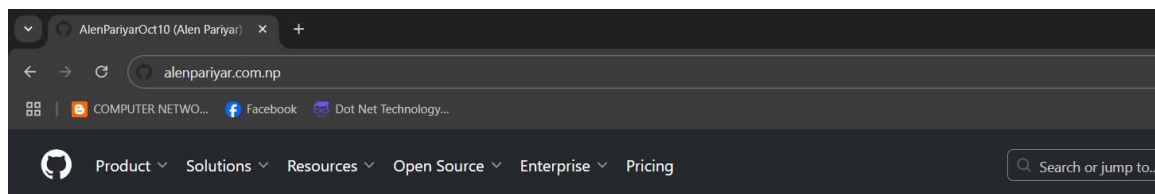


```
C:\Users\oct10>ping alenpariyar.com.np
```

Pinging alenpariyar.com.np [185.27.134.95] with 32 bytes of data:

Step

Step 5 : Visit a website,



Step 6 : Filter packets by IP address,

Capturing from WiFi

File Edit View Go Capture Analyze Statistics Telephony Wireless

ip.host == 185.27.134.95

Source	Destination
192.168.1.68	185.27.134.95
185.27.134.95	192.168.1.68
192.168.1.68	185.27.134.95
185.27.134.95	192.168.1.68
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95
192.168.1.68	185.27.134.95

Step 7 : View Packet,

Capturing from WiFi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.host == 185.27.134.95

Source	Destination	Protocol	Length	Info
192.168.1.68	185.27.134.95	TCP	54	54237 → 80 [FIN, ACK] Seq=1 Ack=1 Win=131072 Len=0
185.27.134.95	192.168.1.68	TCP	54	80 → 54238 [FIN, ACK] Seq=271 Ack=496 Win=30720 Len=0
192.168.1.68	185.27.134.95	TCP	54	54238 → 80 [ACK] Seq=496 Ack=272 Win=130816 Len=0
185.27.134.95	192.168.1.68	TCP	54	80 → 54237 [FIN, ACK] Seq=1 Ack=2 Win=29696 Len=0
192.168.1.68	185.27.134.95	TCP	54	54237 → 80 [ACK] Seq=2 Ack=2 Win=131072 Len=0
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=37/9472, ttl=128 (no response found!)
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=38/9728, ttl=128 (no response found!)
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=39/9984, ttl=128 (no response found!)
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=40/10240, ttl=128 (no response found!)
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=41/10496, ttl=128 (no response found!)
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=42/10752, ttl=128 (no response found!)
192.168.1.68	185.27.134.95	ICMP	74	Echo (ping) request id=0x0001, seq=43/11008, ttl=128 (no response found!)

Destination Address: 185.27.134.95
[Stream index: 18]
Transmission Control Protocol, Src Port: 53813, Dst Port:
Hypertext Transfer Protocol
GET / HTTP/1.1\r\nHost: alenpariyar.com.np\r\nConnection: keep-alive\r\nUpgrade-Insecure-Requests: 1\r\nUser-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36\r\nCookie: __test=3aabe91e86223b2194a564efe3fd4500\r\n\r\n

0010 02 16 2f bb 40 00 80 06 c7 bf c0 a8 01 44 b9 1b ..@...D..
0020 86 5f d2 35 00 50 cf 1e ef 7e 3b 2d 74 dc 50 18 .5.P...~;-t:P..
0030 02 00 02 6b 00 00 47 45 54 20 2f 20 48 54 54 50 ..k..GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 61 6c 65 6e /1.1..Ho st: alen
0050 70 61 72 69 79 61 72 2e 63 6f 6d 2e 6e 70 0d 0a pariyar. com.np..
0060 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 Connection: keep
0070 2d 61 6c 69 76 65 0d 0a 55 70 67 72 61 64 65 2d -alive.. Upgrade-
0080 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 74 Insecure -Request
0090 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e 74 s: 1..Us er-Agent
00a0 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 : Mozill a/5.0 (W
00b0 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b 20 indows N T 10.0;
00c0 57 69 6e 36 34 3b 20 78 36 34 29 20 41 70 70 6c Win64; x 64) Appl
00d0 65 57 65 62 4b 69 74 2f 35 33 37 2e 33 36 20 28 eWebKit/ 537.36 (
00e0 4b 48 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b KHTML, l ike Geck
00f0 6f 29 20 43 68 72 6f 6d 65 2f 31 33 31 2e 30 2e o) Chrom e/131.0.

HTTP Host (http.host), 26 bytes

Packets: 65934 - Displayed: 398 (0.6%)

Profile: Default

Step 8 : Analyze the header,

```

v Transmission Control Protocol, Src Port: 53813, Dst Port:
  Source Port: 53813
  Destination Port: 80
  [Stream index: 1334]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 494]
  Sequence Number: 1      (relative sequence number)
  Sequence Number (raw): 3474911102
  [Next Sequence Number: 495      (relative sequence number)]
  Acknowledgment Number: 1      (relative ack number)
  Acknowledgment number (raw): 992834780
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
  Window: 512

```

The TCP header in this packet contains details of a communication between a client and server over port 80 (HTTP). The **Source Port (53813)** and **Destination Port (80)** identify the endpoints of the connection. The **Sequence Number (1)** indicates the position of this packet's data in the overall communication stream, with the next expected sequence being **495**. The **Acknowledgment Number (1)** confirms receipt of prior data. The header is **20 bytes** long, with the **PSH** and **ACK** flags indicating the delivery of urgent data and acknowledgment of received data, respectively. The **Window Size (131072)** signifies the buffer size for data reception, while the **Checksum (0x026b)** is used for error-checking, though it is unverified. This packet includes **494 bytes of data payload**, making it part of an active HTTP data exchange.